Example 2.2.6 (Putnam 1991) Let $\mathbf{A}$ and $\mathbf{B}$ be different $n \times n$ matrices with real entries. If $\mathbf{A}^3 = \mathbf{B}^3$ and $\mathbf{A}^2\mathbf{B} = \mathbf{B}^2\mathbf{A}$, can $\mathbf{A}^2 + \mathbf{B}^2$ be invertible?

Solution: This is the sort of problem that most students shy away from, even those who excelled at linear algebra. But it is really not hard at all when approached with confidence. First of all, we note the given: $\mathbf{A} \neq \mathbf{B}, \mathbf{A}^3 = \mathbf{B}^3$ and $\mathbf{A}^2\mathbf{B} = \mathbf{B}^2\mathbf{A}$. The conclusion is to determine if $\mathbf{C} = \mathbf{A}^2 + \mathbf{B}^2$ is invertible. Either $\mathbf{C}$ is invertible or it is not. How do we show that a matrix is invertible? One way is to show that its determinant is non-zero. That seems difficult, since the matrices are $n \times n$, where n is arbitrary, and the formula for a determinant is very complicated once $n \geq 3$. Another way to show that a matrix $\mathbf{C}$ is invertible is by showing that $\mathbf{C}b_i \neq 0$ for each basis vector $b_1, b_2, \ldots, b_n$. But that is also hard, since we need to find a basis.

The hunt for a penultimate step for invertibility has failed. That's OK; now we think about *non*-invertibility. That turns out to be a little easier: all we need to do is find a *single* non-zero vector v such that $\mathbf{C}v = 0$. Now that is a manageable penultimate step. Will it work? We have no way of knowing, but the strategic ideas of wishful thinking and make it easier demand that we investigate this path.

We need to use the given, start with $\mathbf{C}$, and somehow get zero. Once again, wishful thinking tells us to look at constructions such as $\mathbf{A}^3 - \mathbf{B}^3$, since $\mathbf{A}^3 = \mathbf{B}^3$. Starting with $\mathbf{C} = \mathbf{A}^2 + \mathbf{B}^2$, the most direct approach to getting the cubic terms seems fruitful (recall that matrix multiplication is not commutative, so that $\mathbf{B}^2\mathbf{A}$ is not necessarily equal to $\mathbf{A}\mathbf{B}^2$):

$$(\mathbf{A}^2 + \mathbf{B}^2)(\mathbf{A} - \mathbf{B}) = \mathbf{A}^3 - \mathbf{A}^2\mathbf{B} + \mathbf{B}^2\mathbf{A} - \mathbf{B}^3 = \mathbf{A}^3 - \mathbf{B}^3 + \mathbf{B}^2\mathbf{A} - \mathbf{A}^2\mathbf{B} = 0.$$

Now we are done! Since $\mathbf{A} \neq \mathbf{B}$, the matrix $\mathbf{A} - \mathbf{B} \neq 0$. Therefore there exists a vector u such that $(\mathbf{A} - \mathbf{B})u \neq 0$. Now just set $v = (\mathbf{A} - \mathbf{B})u$ and we have

$$\mathbf{C}v = \Big((\mathbf{A}^2 + \mathbf{B}^2)(\mathbf{A} - \mathbf{B})\Big)\, u = \mathbf{0}u = 0.$$

Thus $\mathbf{A}^2 + \mathbf{B}^2$ is always non-invertible. ■

Example 2.2.7 (Leningrad Mathematical Olympiad 1988) Let $p(x)$ be a polynomial with real coefficients. Prove that if

$$p(x) - p'(x) - p''(x) + p'''(x) \geq 0$$

for every real x, then $p(x) \geq 0$ for every real x.

Partial Solution: If you have never seen a problem of this kind before, it is quite perplexing. What should derivatives have to do with whether a function is non-negative or not? And why is it important that $p(x)$ be a polynomial?

We have to simplify the problem. What is the most difficult part? Obviously, the complicated expression $p(x) - p'(x) - p''(x) + p'''(x)$. **Factoring** is an important algebraic tactic (see page 148). Motivated by the factorization

$$1 - x - x^2 + x^3 = (1 - x)(1 - x^2),$$

we write

$$p(x) - p'(x) - p''(x) + p'''(x) = (p(x) - p''(x)) - (p(x) - p''(x))'.$$

In other words, if we let $q(x) = p(x) - p''(x)$, then

$$p(x) - p'(x) - p''(x) + p'''(x) = q(x) - q'(x).$$

So now we have a simpler problem to examine:

> *If $q(x)$ is a polynomial and $q(x) - q'(x) \geq 0$ for all real x, what can we say about $q(x)$?*

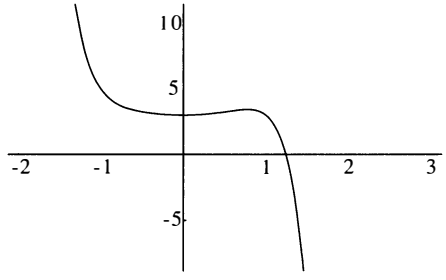
Is it possible as well that $q(x) \geq 0$ for every real x? This may or may not be true, and it may not solve the original problem, but it is certainly worth investigating. Wishful thinking demands that we look into this.

The inequality $q(x) - q'(x) \geq 0$ is equivalent to $q'(x) \leq q(x)$. Consequently, if $q(x) < 0$, then $q'(x)$ must also be negative. Thus, if the graph of $y = q(x)$ ever drops below the x-axis (going from left to right), then it must stay below the x-axis, for the function $q(x)$ will always be decreasing! We have three cases.

- The graph of $y = q(x)$ does cross the x-axis. By the above reasoning, it must only cross once, going from positive to negative (since once it is negative, it stays negative). Furthermore, since $q(x)$ *is a polynomial*, we know that

$$\lim_{x \to -\infty} q(x) = +\infty \quad \text{and} \quad \lim_{x \to +\infty} q(x) = -\infty,$$

because any polynomial $q(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0$ is dominated by its highest-degree term $a_n x^x$ for large enough (positive or negative) x. Therefore, $q(x)$ must have odd degree n and $a_n < 0$. For example, the graph of the polynomial $q(x) = -x^7 + x^2 + 3$ has the appropriate behavior.



However, this polynomial does not satisfy the inequality $q'(x) \leq q(x)$: We have $q(x) = -x^7 + x^2 + 3$ and $q'(x) = -7x^6 + 2x$. Both polynomials are dominated by their highest-degree term. When x is a large positive number, both $q(x)$ and $q'(x)$ will be negative, but $q(x)$ will be larger in absolute value, since its dominant term is degree 7 while the dominant term of $q'(x)$ is degree 6. In other words, for large enough positive x, we will have $q(x) < q'(x)$. Certainly this argument is a general one: if the graph of $y = q(x)$ crosses the x-axis, then the inequality $q'(x) \leq q(x)$ will not be true for all x. So this case is impossible.

- The graph of $y = q(x)$ stays below (or just touches) the x-axis. Then, since $q(x)$ is a polynomial, it must have even degree and a negative leading coefficient. For example, $q(x) = -5x^8 - 200$ would have the right kind of graph. However, the previous argument still applies: for large enough positive x, we will have $q(x) < q'(x)$. So this case is not possible, either.
- The graph of $y = q(x)$ stays above (or just touches) the x-axis, i.e., $q(x) \geq 0$. This case must be true, since we have eliminated the other possibilities! However, it is instructive to see why the previous argument doesn't lead to a contradiction. Now $q(x)$ must have even degree with leading coefficient *positive*, for example, $q(x) = x^2 + 10$ has the right kind of graph. But now, $q'(x) = 2x$. When x is a large positive number, $q(x) > q'(x)$ because the leading coefficients are positive. That's the key.

Anyway, we've managed to prove a very nice assertion:

> *If $q(x)$ is a polynomial with real coefficients satisfying $q(x) \geq q'(x)$ for all real x, then $q(x)$ has even degree with a positive leading coefficient, and is always nonnegative.*

This fact should give us confidence for wrapping up the original problem. We know that $q(x) = p(x) - p''(x)$ has even degree with positive leading coefficient, hence the same is true of $p(x)$. So we have reduced the original problem to a seemingly easier one:

> *Prove that if $p(x)$ has even degree with positive leading coefficient, and $p(x) - p''(x) \geq 0$ for all real x, then $p(x) \geq 0$ for all real x.*

Example 2.2.8 (Putnam 1990) Find all real-valued continuously differentiable functions f on the real line such that for all x,

$$(f(x))^2 = \int_0^x [(f(t))^2 + (f'(t))^2]\,dt + 1990.$$

Partial Solution: What is the worst thing about this problem? It contains both differentiation *and* integration. Differential equations are bad enough, but integral-differential equations are worse! So the strategy is obvious: **make it easier** by differentiating both sides of the equation with respect to x:

$$\frac{d}{dx}(f(x))^2 = \frac{d}{dx}\left(\int_0^x [(f(t))^2 + (f'(t))^2]\,dt + 1990\right).$$

The left-hand side is just $2f(x)f'(x)$ (by the Chain Rule), and the right-hand side becomes $(f(x))^2 + (f'(x))^2$ (the derivative of the constant 1990 vanishes).

Now we have reduced the problem to a differential equation,

$$2f(x)f'(x) = (f(x))^2 + (f'(x))^2.$$

This isn't pretty (yet), but is much nicer than what we started with. Do you see what to do next?

Problems and Exercises

For most of these, your task is just to experiment and get your hands dirty and come up with conjectures. Do not worry about proving your conjectures at this point. The idea is to stay loose and uninhibited, to get used to brainstorming. Some of the questions ask you to conjecture a formula or an **algorithm**. By the latter, we mean a computational procedure that is not a simple formula, but nevertheless is fairly easy to explain and carry out. For example, $f(n) = \sqrt{n!}$ is a formula but the following is an algorithm:

> Compute the sum of every third digit of the base-3 expansion of n. If the sum is even, that is $f(n)$. Otherwise, square it, and that is $f(n)$.

We will return to many of the problems later and develop more rigorous proofs. But it is important that you get your hands dirty *now* and start thinking about them. We reiterate: it is not important at this time to actually "solve" the problems. Having a bunch of partially solved, possibly true conjectures at the back of your mind is not only OK, it is ideal. "Backburner" problems ferment happily, intoxicating your brain with ideas. Some of this fermentation is conscious, some is not. Some ideas will work, others will fail. The more ideas, the better! (By the way, a few of the problems were deliberately chosen to be similar to some of the examples. Remember, one of your orientation strategies is to ask, "Is there a similar problem?")

2.2.9 Define $f(x) = 1/(1-x)$ and denote r iterations of the function f by f^r [see equation (2) on page 32]. Compute $f^{1999}(2000)$.

2.2.10 (Putnam 1990) Let

$$T_0 = 2, \quad T_1 = 3, \quad T_2 = 6,$$

and for $n \geq 3$,

$$T_n = (n+4)T_{n-1} - 4nT_{n-2} + (4n-8)T_{n-3}.$$

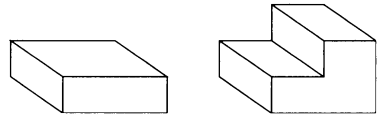
The first few terms are

$$2, 3, 6, 14, 40, 152, 784, 5168, 40576, 363392.$$

Find a formula for T_n of the form $T_n = A_n + B_n$, where (A_n) and (B_n) are well-known sequences.

2.2.11 Let $\mathbb{N}$ denote the natural numbers $\{1, 2, 3, 4, \ldots\}$. Consider a function f that satisfies $f(1) = 1$, $f(2n) = f(n)$ and $f(2n+1) = f(2n) + 1$ for all $n \in \mathbb{N}$. Find a nice simple algorithm for $f(n)$. Your algorithm should be a single sentence long, at most.

2.2.12 Look at, draw, or build several (at least eight) **polyhedra**, i.e., solids with polygonal faces. Below are two examples: a box and a three-dimensional "ell" shape. For each polyhedron, count the number of vertices (corners), faces and edges. For example, the box has eight vertices, six faces and 12 edges, while the ell has 12 vertices, eight faces and 18 edges. Find a pattern and conjecture a rule that connects these three numbers.



2.2.13 Into how many regions is the plane divided by n lines in **general position** (no two lines parallel; no three lines meet in a point)?

2.2.14 A **great circle** is a circle drawn on a sphere that is an "equator;" i.e., its center is also the center of the sphere. There are n great circles on a sphere, no three of which meet at any point. They divide the sphere into how many regions?

2.2.15 For each integer $n > 1$, find *distinct* positive integers x and y such that

$$\frac{1}{x} + \frac{1}{y} = \frac{1}{n}.$$

2.2.16 For each positive integer n, find positive integer solutions $x_1, x_2, \ldots, x_n$ to the equation

$$\frac{1}{x_1} + \frac{1}{x_2} + \cdots + \frac{1}{x_n} + \frac{1}{x_1 x_2 \cdots x_n} = 1.$$

2.2.17 Consider a triangle drawn on the coordinate plane, all of whose vertices are **lattice points** (points

with integer coordinates). Let A, B and I respectively denote the area, number of boundary lattice points and number of interior lattice points of this triangle. For example, the triangle with vertices at $(0,0),(2,0),(1,2)$ has (verify!) $A = 2, B = 4, I = 1$. Can you find a simple relationship between A, B and I that holds for any triangle with vertices at lattice points?

2.2.18 (British Mathematical Olympiad 1996) Define

$$q(n) = \left\lfloor \frac{n}{\lfloor \sqrt{n} \rfloor} \right\rfloor \quad (n = 1,2,\ldots).$$

Determine (with proof) all positive integers n for which $q(n) > q(n+1)$.

2.2.19 Bay Area Rapid Food sells chicken nuggets. You can buy packages of seven or packages of 11. What is the largest integer n such that there is no way to buy exactly n nuggets? Can you generalize this?

2.2.20 Let n be a positive integer.

1. Write down all ordered pairs of integers (a,b), where
   - $1 \le a < b \le n$;
   - $a + b > n$;
   - a and b are relatively prime (share no common divisor besides 1).
2. For each ordered pair (a,b), compute reciprocal of the product of the two numbers, in other words, compute $1/ab$.
3. Add up all of these fractions.

For example, if $n = 6$, the ordered pairs are

$$(1,6),(5,6),(2,5),(3,5),(4,5),(3,4),$$

and the corresponding sum is

$$\frac{1}{6} + \frac{1}{30} + \frac{1}{10} + \frac{1}{15} + \frac{1}{20} + \frac{1}{12} = \frac{1}{2}.$$

Investigate what happens with other values of n, and conjecture something.

2.2.21

(a) Find a nice simple formula for

$$1 + 2 + 3 + \cdots + n,$$

where n is any positive integer.

(b) Find a nice simple formula for

$$1^3 + 2^3 + 3^3 + \cdots + n^3,$$

where n is any positive integer.

(c) Experiment and conjecture the generalization to the above: For each positive integer k, is there a nice formula for

$$1^k + 2^k + 3^k + \cdots + n^k?$$

2.2.22 Define $s(n)$ to be the number of ways in which the positive integer n can be written as an ordered sum of at least one positive integer. For example,

$$4 = 1+3 = 3+1 = 2+2 = 1+1+2$$
$$= 1+2+1 = 2+1+1 = 1+1+1+1,$$

so $s(4) = 8$. Conjecture a general formula.

2.2.23 Find infinitely many positive integer solutions to the equation

$$x^2 + y^2 + z^2 = w^2.$$

2.2.24 Note that

$$6 = 1^2 - 2^2 + 3^2$$

and

$$7 = -1^2 + 2^2 + 3^2 - 4^2 - 5^2 + 6^2.$$

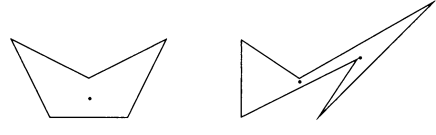
Investigate, generalize, conjecture.

2.2.25 (Turkey 1996) Let

$$\prod_{n=1}^{1996} \left(1 + nx^{3^n}\right) = 1 + a_1 x^{k_1} + a_2 x^{k_2} + \ldots + a_m x^{k_m},$$

where $a_1, a_2, \ldots, a_m$ are nonzero and $k_1 < k_2 < \ldots < k_m$. Find a_{1996}.

2.2.26 *The Gallery Problem.* The walls of a museum gallery form a polygon with n sides, not necessarily regular or even convex. Guards are placed at fixed locations inside the gallery. Assuming that the guards can turn their heads, but do not walk around, what is the minimum number of guards needed to assure that every inch of wall can be observed? When $n = 3$ or $n = 4$, it is obvious that just one guard suffices. This also is true for $n = 5$, although it takes a few pictures to get convinced (the non-convex case is the interesting one). But for $n = 6$, one can create galleries that require two guards. Here are pictures of the $n = 5$ and $n = 6$ cases. The dots indicate positions of guards.



Can you discover a general formula for the number of guards, as a function of n?

2.2.27 *The Josephus Problem.* A group of n people are standing in a circle, numbered consecutively clockwise from 1 to n. Starting with person #2, we remove every other person, proceeding clockwise. For example, if $n = 6$, the people are removed in the order $2,4,6,3,1$, and the last person remaining is #5. Let $j(n)$ denote the last person remaining.

(a) Compute $j(n)$ for $n = 2,3,\ldots 25$.

(b) Find a way to compute $j(n)$ for any positive integer $n > 1$. You may not get a "nice" formula, but try to find a convenient algorithm that is easy to compute by hand or machine.

2.2.28 Let $g(n)$ be the number of odd terms in the row of Pascal's Triangle that starts with $1,n\ldots$. For example, $g(6) = 4$, since the row

$$1,6,15,20,15,6,1$$

contains four odd numbers. Conjecture a formula (or an easy way of computing) $g(n)$.

2.2.29 (Putnam 1991) For each integer $n \geq 0$, let $S(n) = n - m^2$, where m is the greatest integer with $m^2 \leq n$. Define a sequence $(a_k)_{k=0}^{\infty}$ by $a_0 = A$ and $a_{k+1} = a_k + S(a_k)$ for $k \geq 0$. For what positive integers A is this sequence eventually constant?

2.2.30 Complete the solution started in Example 2.2.4.

2.2.31 Complete the solution started in Example 2.2.5.

2.2.32 Let $\{x\}$ denote the closest integer to the real number x. For example, $\{3.1\} = 3$ and $\{4.7\} = 5$. Now define $f(n) := n + \{\sqrt{n}\}$. Prove that, for every positive integer m, the sequence

$$f(m), f(f(m)), f(f(f(m))), \ldots$$

never contains the square of an integer. (Compare this with Example 2.2.5 on page 31.)

2.2.33 Complete the solution started in Example 2.2.7 on page 34.

2.2.34 Complete the solution started in Example 2.2.8 on page 36.

2.2.35 *Cautionary Tales.* It is easy to be seduced by the ease of experimentation-conjecture. But this is only part of mathematical investigation. Sometimes a relatively uninformed investigation leads us astray. Here are two examples. There are many other examples like this; see [17] for a wonderful discussion.

(a) Let $f(n) := n^2 + n + 41$. Is $f(n)$ a prime for all positive integers n?

(b) Let $t(n)$ be the maximum number of different areas that you can divide a circle into when you place n points on the circumference and draw all the possible line segments connecting the points. It is easy to check (verify!) that

$$t(1) = 1, t(2) = 2, t(3) = 4, t(4) = 8, t(5) = 16.$$

The conjecture that $t(n) = 2^{n-1}$ is practically inescapable. Yet $t(6)$ is equal to 31, not 32 (again, verify!), so something else is going on. Anyway, can you deduce the correct formula for $t(n)$?

2.3 Methods of Argument

As we said earlier, the solution to every problem involves two parts: the investigation, during which you discover what is going on, and the argument, in which you convince others (or maybe just yourself!) of your discoveries. Your initial investigation may suggest a tentative method of argument. Of course, sometimes a problem divides into cases or sub-problems, each of which may require completely different methods of argument.

Arguments should be rigorous and clear. However, "rigor" and "clarity" are both subjective terms. Certainly, you should avoid glaring logical flaws or gaps in your reasoning. This is easier said than done, of course. The more complicated the argument, the harder it is to decide if it is logically correct. Likewise, you should avoid deliberately vague statements, of course, but the ultimate clarity of your argument depends